



Testing stationarity of the detrended price return in stock markets

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ABSTRACT

This paper proposes a generalized porous media equation with drift as the governing equation for stock market indexes. The proposed governing equation can be expressed as a Fokker–Planck equation (FPE) with a non-constant diffusion coefficient. The governing equation accounts for non-stationary effects and describes the time evolution of the probability distribution function (PDF) of the price return. By applying Ito's Lemma, the FPE is associated with a stochastic differential equation (SDE) that models the time evolution of the price return in a fashion different from the classical Black–Scholes equation. Both FPE and SDE equations account for a deterministic part or trend and a stochastic part or q-Gaussian noise. The q-Gaussian noise can be decomposed into a Gaussian noise affected by a standard deviation or volatility. The presented model is validated using the S&P500 index's data from the past 25 years per minute. We show that the price return becomes Gaussian, consequently stationary by normalizing the detrended data set. The normalization of the data is calculated by subtracting the trend and then dividing by the standard deviation of the detrended price return. The stationarity test consists of representing the power spectrum in terms of the time series's autocorrelation. Additionally, this paper presents the multifractal analysis for the detrended and normalized price return to describe the Hurst exponent dynamics over the dataset.

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1. Introduction

Complex dynamic systems encountered in business [1], financial markets [2], genetics [3,4], neuroscience [5,6], biomedicine [7,8], ocean dynamics [9] and seismology [10] generate non-stationary time series. A time series is non-stationarity when it is non-identically distributed throughout their full length of time [11,12]. In stock markets, these type of time series are simple price return [2,12–14], volatility of index price [15] and stock market index [16,17]. In the statistical analysis of financial markets, the non-stationarity time series that is extensively analyzed is the simple price return, which is an arithmetical difference [17]. For price return, the non-stationarity is produced by the non-constant activity during a trading day and the heterogeneity of market participants [2,18].

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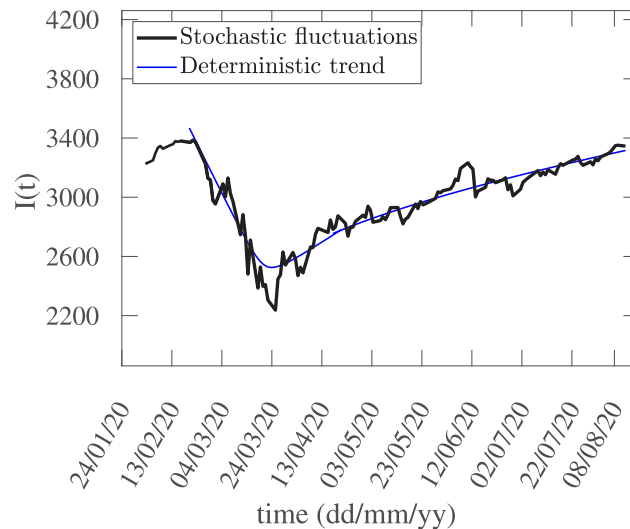


Fig. 1. Schematic plot of a downward and upward swing in the S&P500. The index is decomposed into a deterministic trend (blue) and stochastic fluctuations (black).

The non-stationary time series are difficult to analyze because the process remains in a non-equilibrium state. Consequently, their models do not achieve optimal forecasting and control [1,19]. Therefore, it is challenging to obtain an accurate model that can be used to calculate the probability of future value based on initial conditions.

In recent years, extensive research has been devoted to model the index price return distributions considering its statistical features, and non-stationarity characteristics [20–22]. The Black–Scholes equation have been the governing equation to describe the characteristics of the stock return distribution [23–25]. This important model used the Geometric Brownian Motion (GBM) to model the index price return. However, it fails to capture the non-Gaussian properties of the price return [25], negative skewness [21], and negative values obtained from recent events [26]. As a result, different models started to appear, applying modifications or generalizations to the GBM. The Fractional Brownian Motion (FBM) [27], the Mixed Fractional Brownian Motion (MFBM) as a combination of the Brownian motion with the FBM [28], the Generalized Mixed Fractional Brownian Motion (GMFBM) [29,30], and Fokker–Planck Equation [31]. These new models capture better the characteristic features of the index price return. However, some of them were developed considering a neutral risk assumption [29,32]. As an alternative to these traditional stock return models, an increasing number of quantum models have also been applied to study the stochastic dynamics of stock prices. The most well-known are; the Quantum Spatial-periodic Harmonic model (QSH) [20], the Quantum Brownian Motion model (QBM) [22], and the Quantum Harmonic Oscillator model (QHO) [21]. An advantage of these models is that they incorporate the market uncertainty (risk), although they have been derived from the stochastic equation of a Brownian process.

The contribution of this paper is the presentation of a simpler but more effective governing equation to model the non-stationarity of price return. This governing equation is constructed based on the decomposition of the time-series by a deterministic part or trend and a stochastic part or q-Gaussian noise. As an example, the Fig. 1 shows the index fluctuations of the S&P500 during the initial COVID-19 outbreak, where the stock market index can be decomposed into a deterministic trend and stochastic fluctuations. The stochastic part adequately represents the stylized facts of the financial time series such as the fat-tailed distribution, aggregational normality and self-similarity with finite variance. Therefore, the stochastic part can be modeled by a self-similar q-Gaussian distribution centered on zero. To validate our governing equation, the trend of the S&P500 index data is removed by applying a moving average method (MA) based on the Kurtosis evaluation. By applying an arithmetical difference to the detrended S&P500 index data, the detrended price return is calculated. The detrended price return exhibits fat tails, short-term correlation and volatility clustering. The volatility clustering effect represents a non-constant standard deviation over the full length of the time series [19]. As a consequence, the detrended price return is non-identically distributed, and therefore, non-stationary. To remove the volatility clusters, we divided the detrended price return by the standard deviation, i.e. it was “standardized”. This process is called “normalization of the data set”. We show that the normalized price return is a stationary process by applying the power spectral density theory. Hence, the detrended and normalized price return is identically distributed throughout their full length of time holding finite and time-invariant values of mean and variance.

Additionally, this paper presents the scaling analysis of the detrended price return and the normalized price return by estimating the Hurst exponent H [33]. Time series are classified as monofractal or multifractal depending on how the Hurst exponent varies over several orders of magnitude [34]. A time series is called monofractal if the Hurst exponent is constant for all the orders of magnitude, otherwise is called multifractal [35].

This paper is divided into four parts: In Section 2, the governing equation of the simple price return is presented. Section 3 presents a straightforward way to detrend the stock market price, based on the optimization of a moving window via the kurtosis evaluation. Then, the normalized price return is obtained right after dividing the detrended price return by the standard deviation. In Section 4, we perform the tests on the normalized price return to show that it is stationary. The test verifies that the power spectrum of the time series equals the Fourier transform of its autocorrelation [36,37]. Section 5, is based on the evaluation of the Hurst exponent in terms of the orders of the statistical moments of the time series. We provide the details of the Hurst exponent calculation and the characterization of the detrended price return. Finally, the conclusions and perspectives of this work are presented in Section 6.

2. Governing equation of price return

The conventional approach to option pricing analysis leads to the classical Black–Scholes equation or non-linear Black–Scholes equation (BSE) as the governing equation of price return [38]. A limitation of the BSE is the assumption that stock prices follow a geometric Brownian motion pattern, ignoring that prices can have sustained trends and even can reach negative values in the real indexes [26]. Our governing equation covers BSEs' limitations, allowing the price to fluctuates in a particular direction (trend) and incorporates stochastic or q-Gaussian noise. The q-Gaussian noise is a better descriptor of the fluctuations in stock markets [39]. For this analysis, the price return at time t is defined as [40–42]:

$$X(t_0, t) = I(t_0 + t) - I(t_0), \quad (1)$$

where $I(t_0)$ is the stock market index at time t_0 , and $I(t)$ is the stock market index for any time $t > t_0$.

We assume that the stock market index $I(t)$ can be decomposed into a deterministic trend $\bar{I}(t)$ and a stochastic time series $I^*(t)$:

$$I(t) = \bar{I}(t) + I^*(t). \quad (2)$$

Then, the price return $X(t)$ is split into a deterministic $\bar{X}(t)$ component and stochastic or q-Gaussian noise $x(t)$:

$$X(t) = \bar{X}(t) + x(t), \quad (3)$$

where $\bar{X}(t) = \bar{I}(t_0 + t) - \bar{I}(t_0)$ and $x(t) = I^*(t_0 + t) - I^*(t)$. This equation can be written as an Stochastic Differential Equation (SDE)

$$dX(t) = d\bar{X}(t) + dx(t), \quad (4)$$

where $\bar{X}(t) = \bar{I}(t + dt) - \bar{I}(t_0)$ and $x(t) = I^*(t + dt) - I^*(t)$, and dt is an infinitesimal time. Our aim is to find the components of this SDE. Based on previous work [39], we postulate that the probability density function (PDF) of the detrended price return $x(t)$ is described by the functional form :

$$P(x, t) = \frac{1}{(Dt)^{1/\alpha}} \left[g_q \left(\frac{x}{(Dt)^{1/\alpha}} \right) \right] \quad (5)$$

where D , q and α are time-dependent fitting exponents, and $g_q(x)$ is the function so-called normalized q-Gauss distribution function. This distribution function is often used in the non-extensive statistical mechanics, because allows the generalization of Boltzmann entropy. The $g_q(x)$ is defined as: [43]:

$$g_q(x) = \frac{1}{C_q} \left[1 - (1 - q)x^2 \right]^{\frac{1}{1-q}}, \quad (6)$$

The normalization constant C_q for $1 < q < 3$ is given by:

$$C_q = \sqrt{\frac{\pi}{q-1}} \frac{\Gamma((3-q)/(2(q-1)))}{\Gamma(1/(q-1))}. \quad (7)$$

Then, we apply the following change of variable $T = t^\xi$, where $\xi = \frac{3-q}{\alpha}$, in the governing equation of $P(x, t)$ [Eq.(12) from [39]]. The corresponding Partial Differential Equation (PDE) is:

$$\frac{\partial P}{\partial T} = D^\xi \frac{\partial^2 P^{2-q}}{\partial x^2}, \quad (8)$$

where the diffusion term is D^ξ . By applying Eqs. (3), (5), and (6), the PDF of price return $P(X, T)$ is:

$$P(X, T) = \frac{1}{(D^\xi T)^{1/(\xi\alpha)}} \frac{1}{C_q} \left[1 - (1 - q) \frac{[X - \bar{X}(T)]^2}{(D^\xi T)^{2/(\xi\alpha)}} \right]^{\frac{1}{1-q}}. \quad (9)$$

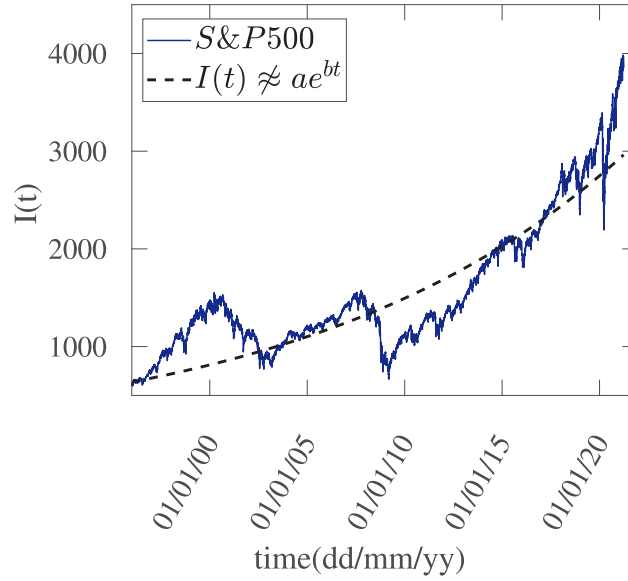


Fig. 2. (Stock market index $I(t)$ of S&P500 in the period of 02/01/96 to 30/03/21 (more than 25 years). Considering the GBM in Eq. (16), the deterministic part is modeled to find the trend of the S&P500. As a result, the fitted line ae^{bt} is an approximation of the S&P500 data trend, where $a = (9.845 \pm 0.7) \times 10^{-51}$ and $b = (1.668 \pm 0.001) \times 10^{-4}$. We note that when this approximated trend is subtracted, the fluctuations will not form a normal distribution function. This rules out the validity of Eq. (16).

Then, the chain rule is applied to obtain the governing equation of $P(X, T)$.

$$\begin{aligned} \frac{\partial^2}{\partial^2 x} &= \frac{\partial^2}{\partial^2 X} \\ \frac{\partial}{\partial T} &= \frac{1}{\xi} t^{1-\xi} \frac{\partial}{\partial t} + \frac{\partial \bar{X}(T)}{\partial T} \frac{\partial}{\partial X} \end{aligned} \quad (10)$$

By replacing Eq(10) in Eq(8), the general PDE can be defined considering that $\frac{t^{1-\xi}}{\xi} = \frac{\partial t}{\partial T}$ as:

$$\frac{\partial P}{\partial T} = D^\xi \frac{\partial^2 P^{2-q}}{\partial X^2} - \frac{\partial \bar{X}(T)}{\partial T} \frac{\partial P}{\partial X}. \quad (11)$$

The term D^ξ is the diffusive parameter, which is independent on X . Consequently, a comparison between Eq(11) with the linear Fokker–Planck equation (LFPE) is made. The LFPE is :

$$\frac{\partial P(X, T)}{\partial \tau} = - \frac{\partial (D_1 P(X, T))}{\partial x} + \frac{\partial^2 (D_2 P(X, T))}{\partial x^2}, \quad (12)$$

After that, the diffusion coefficient in the LFPE is defined as: $D_2 = D^\xi P(X, T)^{1-q}$. By replacing Eq(9) into this equation, we obtain an explicit expression for diffusion coefficient of the LFPE:

$$D_2 = D^{2/\alpha} C_q^{q-1} T^{(q-1)/(3-q)} \left[1 - \frac{(1-q)[X - \bar{X}(T)]^2}{(D^\xi T)^{2/(3-q)}} \right] \quad (13)$$

and the drift term is:

$$D_1 = \frac{\partial \bar{X}(t)}{\partial T} \quad (14)$$

The drift term is the overall direction of the index price showing particular tendencies (trend). The trend is associated with the systematic risk response, where external events can produce a change of trend direction [44]. This feature implies that future price changes are dependent on the past values and therefore it is possible to develop a forecast in the short term with some degree of confidence.

The Fokker–Planck equation in Eq. (12) is used to obtain the probability distribution of price return at any time. The stochastic differential equation (SDE) represents the dynamics of price return as a function of time [45]. The SDE can be obtained by applying the Itô Lemma [45] in Eq. (12):

$$dX_T = \mu(T)dT + \sigma(X, T)dW_T, \quad (15)$$

with drift $\mu(T) = D_1(T)$ and variance $\sigma^2(X, T) = 2D_2(X, T)$. The function W_T is the Gaussian noise, and the term $\sigma(X, T)dW_T$ corresponds to q-Gaussian noise. The interpretation of our SDE is that in a small time interval, dT , the price return changes its value due two effects. The first one is related with the drift D_1 that depends on time only. The second is the diffusion term D_2 that depends on time and price return.

The SDE in Eq. (15) is different from the Geometrical Brownian Motion (GBM) [46]:

$$dI_T = \mu I_T dT + \sigma I_T dB_T, \quad (16)$$

where, μ and σ values are constants, and B_T is the standard Brownian noise. The GBM is the basic model for stock price dynamics in the Black–Scholes framework [24,45]. If we consider the approximation, $Z_T = dI_T/I_T$, where Z_T is a logarithmic difference of the stock market index $Z(t_0, t) = \ln \left[\frac{I(t_0+t)}{I(t_0)} \right]$. Then, Eq. (16) can be simplified as $Z_T = \mu dT + \sigma dB_T$. The solution of this equation is an exponential growth superposed by a normally distributed Brownian process [24,45]. This solution constitutes a rough approximation of the price return, as shown in Fig. 2. In some studies, the log-return Z_T is often used for technical analysis because it is close to the percentage price change [47–49], a concept often used to neutralize most of the non-stationary effects [50–52]. One of the key properties of the GBM approach to modeling asset prices is that prices will not go negative, an assumption that may not be reasonable given recent events in some asset markets. Some examples are crude oil's price fall and the bankruptcy of some assets of global energy companies, which had reached negative values [26]. Additionally, in the GBM, it is assumed that the natural logarithmic of price return follows the Brownian motion with drift and that the price return has a normal distribution [45]. However, price returns or log price returns have been modeled by a q-Gaussian distribution function [53]. Thus, the stochastic differential equation, Eq. (15), could potentially model this behavior better than the standard GBM.

3. Detrending time series

In the previous section, we postulated that the time series of the price return can be decomposed into a deterministic time series and q-Gaussian noise. In what follows, we validate this hypothesis by performing an analysis of the S&P500 index. The analyzed data corresponds to a time frame from January 1st, 1996 to March 31th, 2021 with a frequency of one minute. The S&P500 stock market index data was obtained from the Thomson Reuters Corporation [54]. Before the analysis, few artifacts were removed from the data [55–57], as explained in Appendix A.

The Moving Average is a widely known method to remove the trend in time series [3,39,58]. Many variations and implementations have been developed by researchers, but its underlying purpose remains, which is to determine the trend. Despite new models, the MA method is still considered as a good option for detrending the data due to its easiness, objectiveness, reliability, and usefulness [58].

The MA is a common average within a time window size that is shifted forward until the end of the data set [58]. Each point in the time series data is equally weighted. The arithmetic centered moving average has been the most widely used MA case [3].

The MA method offers different options for its calculation. The first option is regarding how the time window will be shifted until the end of the data. For this analysis, it has been used an overlapped or rolling window. These windows are extended over an interval $[t, t + t_w]$, where the consecutive window is incremented by one time point to: $[t + 1, t + t_w + 1]$. The t_w represents the size of the time window. The second option is choosing the polynomial order k for fitting each time segment for the detrending process. The original MA uses $k = 0$, which represents an unweighted mean [59]. The option $k = 1$ is used in other studies when the focus is on the local trend (particular tendency per window) [4]. These options were tested and applied in the index price $I(t)$. In our analysis, there is no major difference between $k = 0$ and $k = 1$, so $k = 0$ was used.

3.1. Optimal time window

The size of the time window, t_w , is the key parameter to obtain stationary detrended data. The distribution of the data set of each window t_w should be self-similar, without outliers (extreme events), but with time-dependent standard deviation. Xu et al. [53] applied the kurtosis definition to compare each distribution of the data set of each window t_w . The kurtosis of a time series X is defined as the fourth standardized moment, defined as the relation between central moments $\langle u^n \rangle$ of order n , where $u = X - \langle X \rangle$,

$$K = \frac{\langle u^4 \rangle}{\langle u^2 \rangle^2}, \quad (17)$$

where the delimiters $\langle \rangle$ denotes time average for each particular time window.

The optimal time window should satisfy the condition of Kurtosis equals three to obtain an optimum time window size such that for a given data set, the distribution in each window is as close as possible to a Gaussian distribution [60]. Additionally, a Kurtosis equals three implies a time series in which extreme or outlier events are improbable to occur, with a skewness equals to zero. The skewness is defined as a normalized third-order moment of the PDF. As a consequence, the time series remains symmetric around its mean value [61].

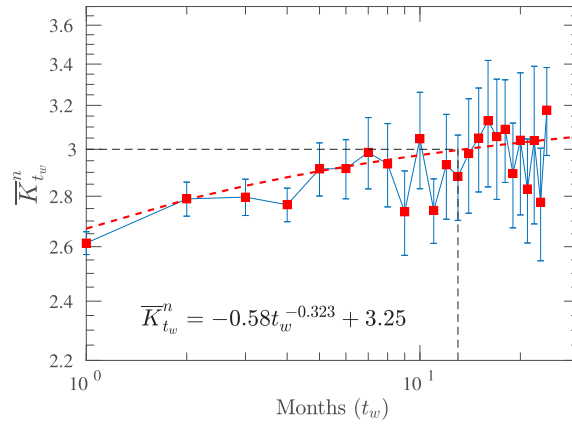


Fig. 3. Optimal window size t_w that will be used for the Moving Average. The $\bar{K}_{t_w}^n = 3$ yields a time window of 13 ± 1 months. The smaller time window, t_w , is chosen to preserve data features that tend to attenuate with longer time windows. The blue bars are graphical representations of the standard error of $\bar{K}_{t_w}^n$ data in average per t_w .

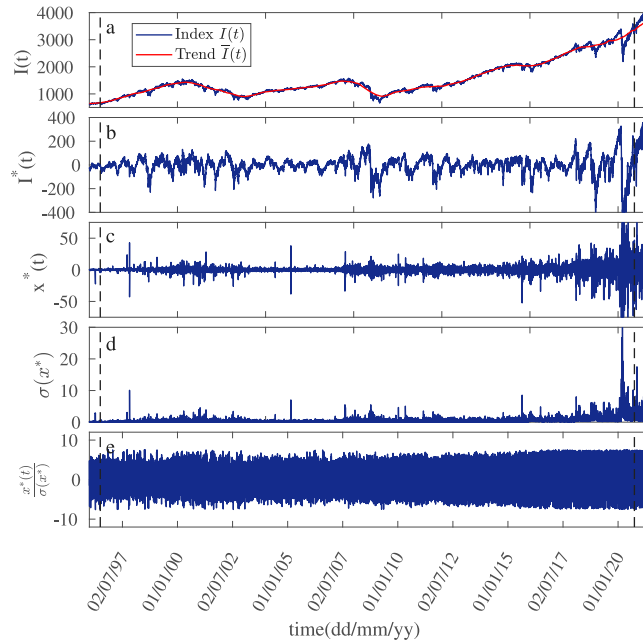


Fig. 4. (a) Trend obtained after applied the MA analysis for a time window of 12 months for S&P500 data. (b) Detrend price $I^*(t)$ after subtracting the trend shown on subfigure 4-a. (c) Detrended price return, $x^*(t)$, obtained from the differences between adjacent elements of $I^*(t)$. (d) The standard deviation is obtained from an overlapping or rolling window of 1 h. Within a time period of 1 h, the time-dependence of the standard deviation is captured before it dissipates. (e) The subfigure represents the normalized price return. The normalization of the detrended price return is calculated by dividing the data set by the standard deviation previously calculated as $\sigma(x^*)$.

Before applying the MA method, we developed a preliminary process to choose an optimal time window t_w . We used a range of values between 1 to 25 months as a possible time window. For each possible t_w value, the total length of N of the time series is split into non-overlapping time windows. The total length of N is often not a multiple of the time window, t_w . Consequently, the number of segments is defined by rounding them to the nearest lower integer $\lfloor n = N/t_w \rfloor$. Next, the kurtosis in each j th window is calculated, where $j = 1, 2, 3, 4, \dots, n$. and averaged for the n windows as follows;

$$\bar{K}_{t_w}^n = \frac{1}{n} \sum_{j=1}^n K_{t_w}(j), \quad (18)$$

where $\bar{K}_{t_w}^n$ denotes the average over the n windows. Fig. 3 presents the results of $\bar{K}_{t_w}^n$. The optimal window size is 13 ± 1 months. The time window assumed for the MA analysis is 12 months. The smaller value of time window, t_w , is chosen

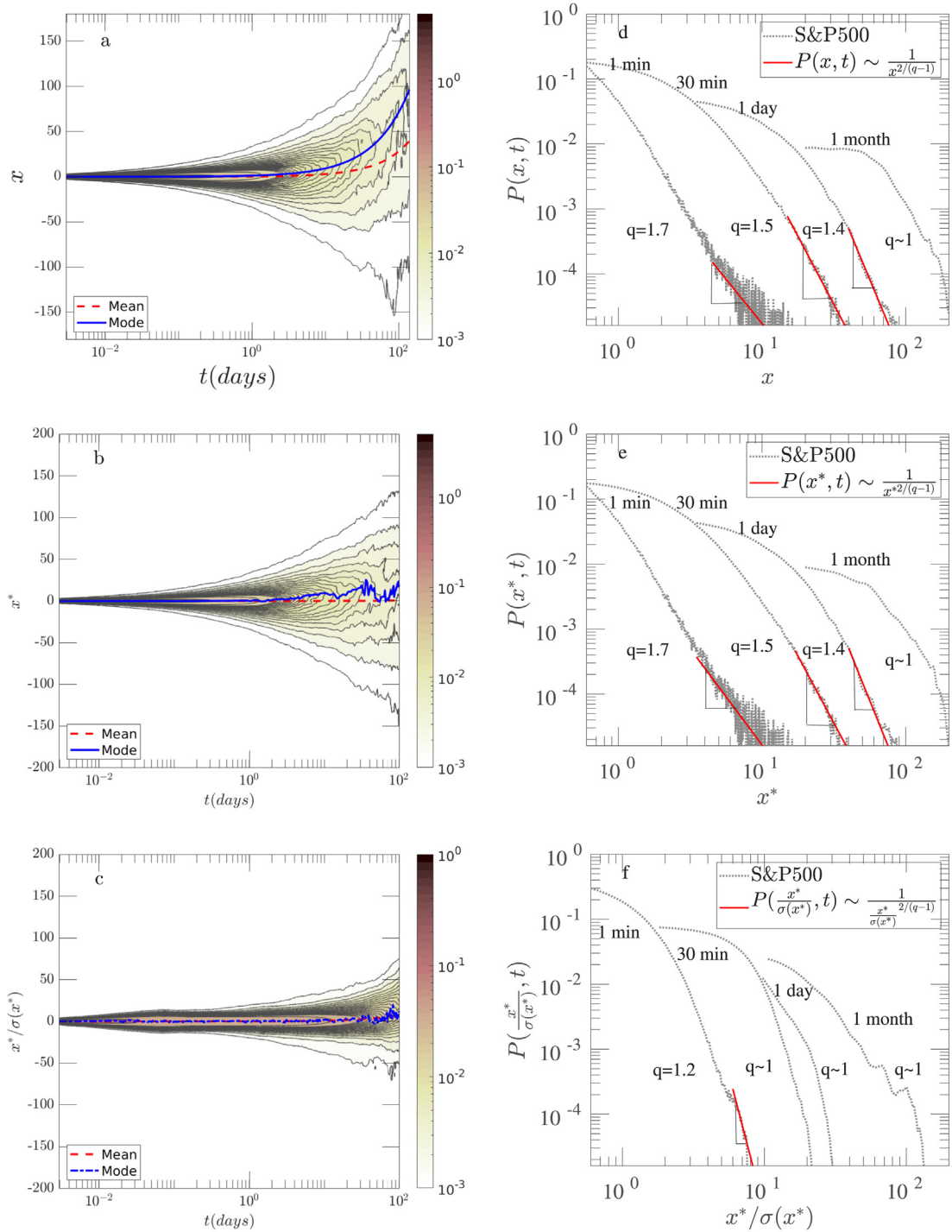


Fig. 5. Representation of the time evolution of the PDF for the different types of price return displaying the asymptotic behavior (tail behavior). The construction of the PDF's evolution was made considering particular times intervals to calculate the price increments or price return. (a) Time evolution of the PDF of price return. (b) Time evolution of the PDF of detrended price return. (c) Time evolution of the PDF of the normalized price return. (d) Asymptotic behavior for the PDFs of price return at particular times. (e) Asymptotic behavior for the PDF of detrended price return at particular times. (f) Asymptotic behavior for the PDF of normalized price return at particular times. A faster convergence to Gaussian, within 30 min., is noted when the normalized price return is used.

because it will preserve features of the data such as peak height and width, which is usually attenuated with a longer time window t_w .

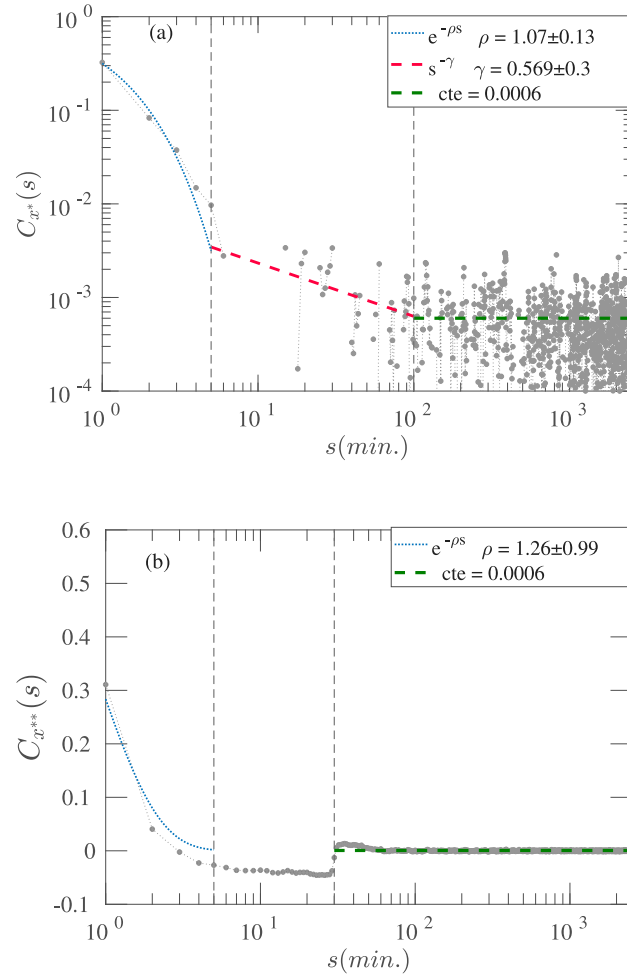


Fig. 6. Autocorrelation of the detrended price return calculated as $C_x(s) \equiv \frac{\langle x_t x_{t+s} \rangle}{\sigma_x^2}$. Short-time correlations are observed for the first 6 min. A weak long-time correlation appears after 100 min.

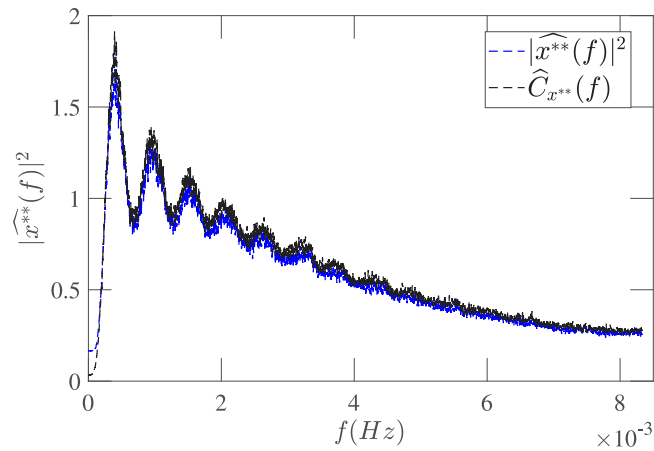


Fig. 7. The Power-spectral density $|\hat{x}(f)|^2$ of the detrended price return matches with the Fourier transform of its autocorrelation..

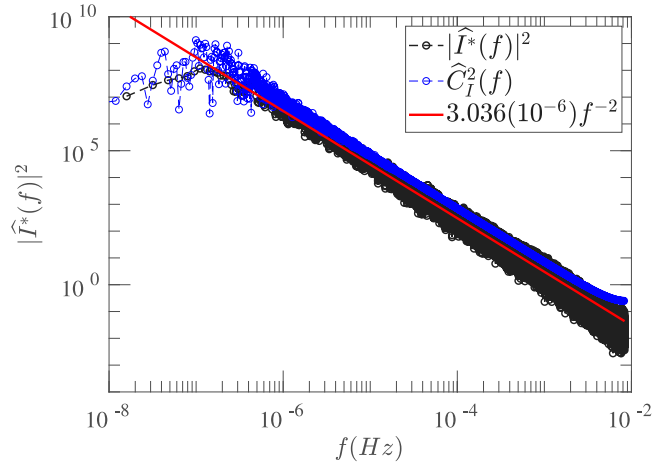


Fig. 8. The power-spectral density of the detrended index $I^*(t)$ holds a power law $P(f) = M/f^2$, where $M = 3.036(10^{-6}) \pm 3(10^{-9})$. The following relation $|\hat{I}(f)|^2 = \hat{C}_I^2(f)$ is observed, suggesting that the Index once detrended is stationary as well.

3.2. Calculation of trend and fluctuating part

The non-stationary time series of the stock market index $I(t)$ is decomposed into a trend $\bar{I}(t)$ and a stochastic part $I^*(t)$,

$$I^*(t) = I(t) - \bar{I}(t). \quad (19)$$

The trend $\bar{I}(t)$ is constructed for the index price of S&P500 applying MA method with $k = 0$ and considering overlapped windows of $t_w = 12$ months. This process is constructed based on a weighted sum or average of each time window. Considering that the time windows for $t < \frac{t_w}{2}$ and $t > N - \frac{t_w}{2}$ are truncated, we identified three cases to calculate the moving average that depends on the position of the moving window [62]:

- For $t < \frac{t_w}{2}$

$$\bar{I}(t) = \frac{2}{t_w + 2t} \sum_{k=-t+1}^{\lceil (t_w-1)/2 \rceil} I(t+k) \quad (20)$$

- For $\frac{t_w}{2} < t < N - \frac{t_w}{2}$

$$\bar{I}(t) = \frac{1}{t_w} \sum_{k=-\lfloor (t_w-1)/2 \rfloor}^{\lceil (t_w-1)/2 \rceil} I(t+k) \quad (21)$$

- For $t > N - \frac{t_w}{2}$

$$\bar{I}(t) = \frac{2}{2N - 2t + t_w} \sum_{k=-\lfloor (t_w-1)/2 \rfloor}^{\lceil N-t \rceil} I(t+k), \quad (22)$$

with the time step of $t = 1, 2, 3, \dots, N$ for the index fluctuations.

Fig. 4-a presents the trend $\bar{I}(t)$ after applying the MA analysis for a time window of 12 months (red line). The trend obtained represents a general tendency of the stock market index $I(t)$. Fig. 4-b shows the detrended price as a result of subtracting the obtained trend.

After the detrended process, the expected result is that the price return will lose asymmetry and will be modeled by the self-similar q-Gaussian distribution function (Eq. (5)). However, the detrended price return is still non-stationary because it presents volatility clusters over the time series's full length (Fig. 4-b). These clusters over the detrended price return represent a time-dependent standard deviation. Considering Eq. (15), the stochastic part or q-Gaussian noise is compound by a Gaussian or stationarity noise affected by a standard deviation that varies with time. Then, the normalized price return is obtained after dividing the detrended price return by its standard deviation to obtained a stationarity time series.

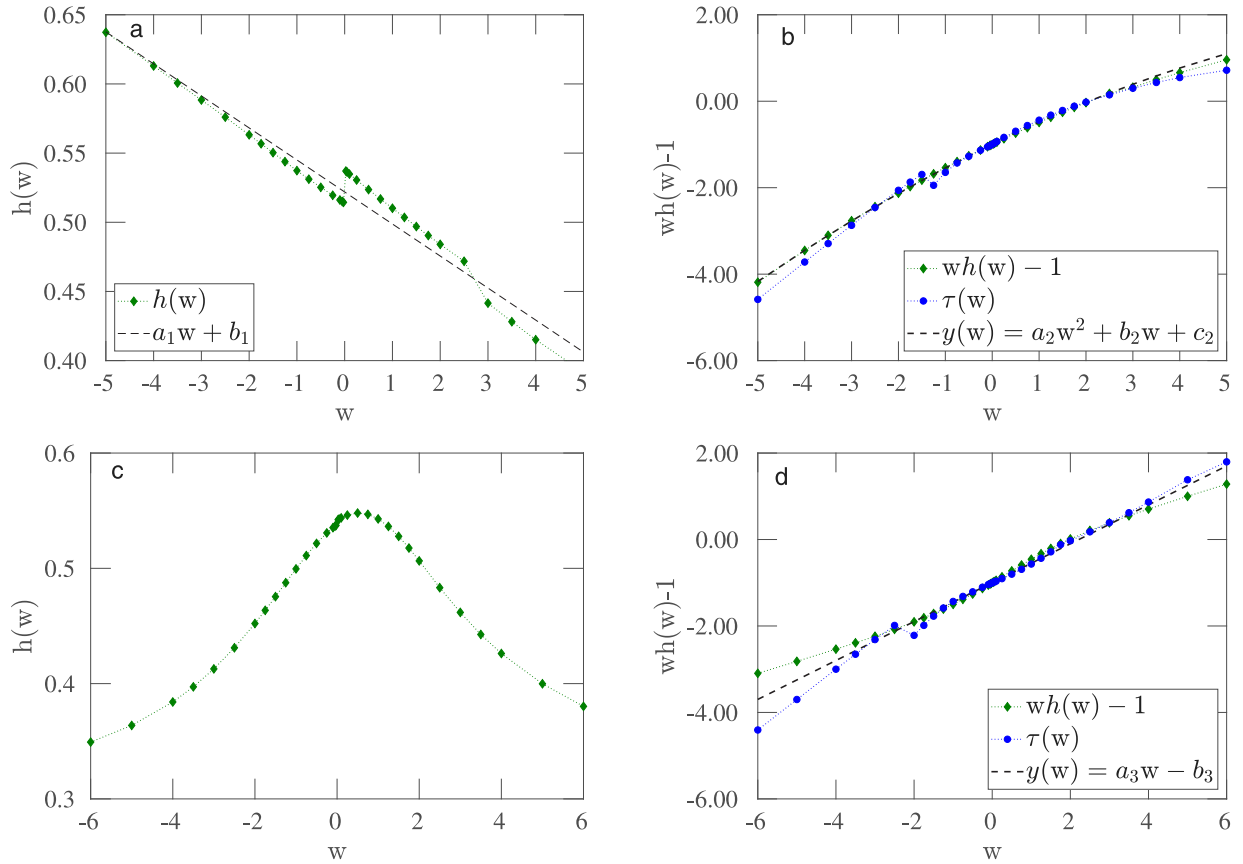


Fig. 9. Evaluation of the scale exponents, $h(w)$, and the generalized scaling exponent, $\tau(w)$ for the detrended and the normalized price return. The subfigures (a) and (b) corresponds to the analysis of detrended price return and are obtained from results shown in Figs. C.1 and C.2 respectively. The subfigures (c) and (d) corresponds to the analysis of normalized price return and are obtained from results shown in Figs. C.3 and C.4 respectively. The generalized Hurst exponent $h(w)$ follows Eq. (28) for a normalized and multifractal time series. (a) For the fitting of the generalized Hurst exponent $a_1 = -0.0232 \pm 0.002$ and $b_1 = 0.5219 \pm 0.004$. (b) For the fitting of the generalized scaling exponent $a_2 = -0.02157 \pm 3.8(10^{-4})$, $b_2 = 0.527 \pm 0.0012$, $c_2 = -0.9954 \pm 0.0034$. (c) For the normalized price return the generalized Hurst exponent obeys a curve bell-shaped which points to 0.35 when $|w| > 6$. (d) For the fitting of the generalized scaling exponent, normalized price return case, $a_3 = 0.45 \pm 0.01$ and $b_3 = 0.999 \pm 0.04$.

The standard deviation is obtained from an overlapping or rolling window of 1 h. Within a time period of 1 h, the highest volume of option trading occurs, consequently the standard deviation is captured before it dissipates [63]. Fig. 4-e shows how the clusters disappears after dividing the detrended price return by its standard deviation. Fig. 5, presents the time evolution of the PDFs of simple price return, detrended price return, and normalized price return, respectively. Fig. 5-a shows that the simple price return has a positive trend and mode that grows with a positive rate proportional to the time interval of the price return. Fig. 5-b shows that the PDF of detrended price return has a constant mean value while its mode oscillates around zero. Fig. 5-c shows that the PDF of normalized price return has constants mean and mode values. The fat-tailed distributions of the PDFs for detrended price return represent the simple price return's stylized facts adequately. For this case, the same rate of aggregation of normality can be observed for the price return with the only difference that it is centered in zero (Figs. 5-d and 5-e). On the other hand, the normalized price return presents a convergence to normal distribution that occurs at a faster rate (Fig. 5-f). The fat-tailed distribution fastly transforms into a Gaussian one for time intervals larger than 30 min.

4. Stationarity of detrended price return

In the following section, we test the detrended time series for stationarity. In statistics one of the most widely applied method to test stationary is the Dickey-Fuller test [64–66]. However, this method is applied for autoregressive time series with normal random noise (ARIMA) [66]. In Appendix-B we shown that the detrended price return can be represented as an autoregressive model but holding a q-Gaussian noise only. As consequence, the Dickey-Fuller test will not be applicable for the stationarity test. The proposed tests in this paper are based on the calculation of the power spectrum analysis [36].

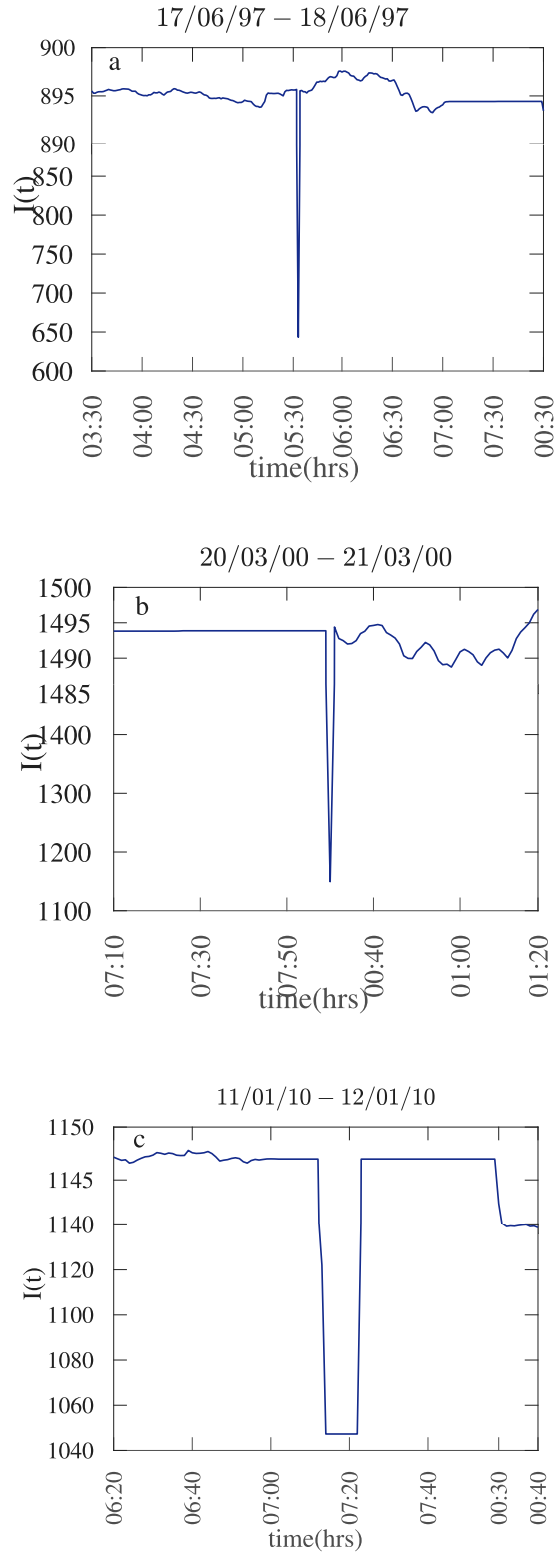


Fig. A.1. The three events that produce spurious jumps in S&P500 data over the past 24 years are shown in subfigures (a) the peak of the Asian financial crisis, (b) The imminent bankruptcy of many Internet companies was announced, and (c) The Haiti earthquake effect.

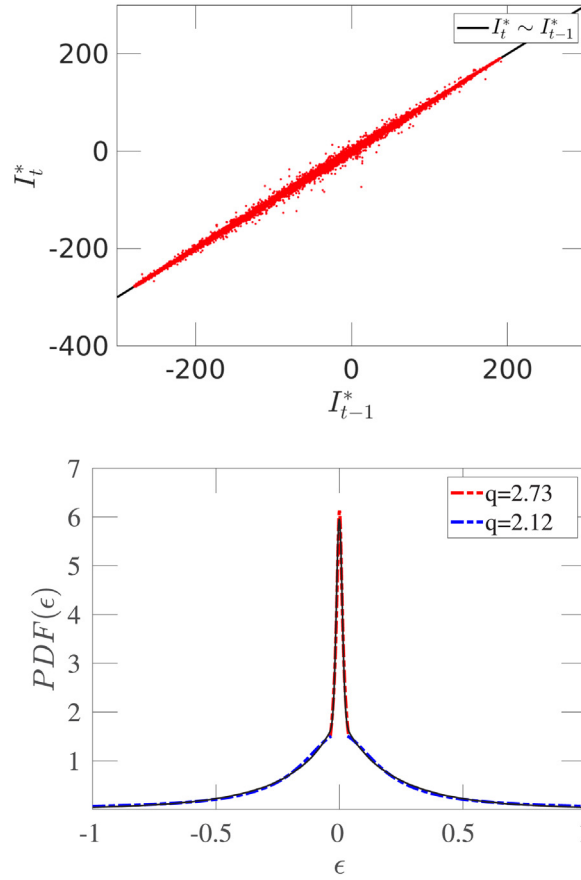


Fig. B.1. (a) Linear relationship between I_t^* and I_{t-1}^* . (b) The noise obtained by removing the linear trend obeys a q -Gaussian distribution.

We recall the expression for the detrended price return:

$$x^*(t) = I^*(t_0 + t) - I^*(t_0), \quad (23)$$

where $I^*(t_0)$ is the detrended stock market index at time t_0 , and $I^*(t)$ is the detrended stock market index for any time $t > t_0$.

Looking back the definition of normalization, we defined the normalized price return as;

$$x^{**} = \frac{x^*(t)}{\sigma(x^*)}, \quad (24)$$

First, we proceed to evaluate the correlation of the detrended price return x^* , and normalized price return x^{**} . The autocorrelation is defined as [67,68];

$$C_x(s) \equiv \frac{\langle x_t x_{t+s} \rangle}{\sigma_x^2}, \quad (25)$$

$$\sigma_x^2 = \langle (x_t - \mu)(x_t - \mu) \rangle.$$

where a particular price return for a specific time is represented by x_t and the time lag by s . The autocorrelation is evaluated for two cases after replacing x by the detrended price return and the normalized price return. For both cases short-time autocorrelations are observed in Fig. 6. Fig. 6-a represents the autocorrelation for the detrended price return. For this case, the power-law fitting of the autocorrelation holds only one order of magnitude. The autocorrelation has an exponential trend for lower values of s , and for higher values, the autocorrelation shows a weak but non-zero autocorrelation. A transition zone is observed between lower and higher time-lag values, and it decreases rapidly with a power law that has an exponent $\gamma = -0.603 \pm 0.32$. Fig. 6-b represents the autocorrelation for the normalized price return. For this case, the autocorrelation has an exponential trend for lower values of s . The autocorrelation presents negative values until the first 30 min of lag.

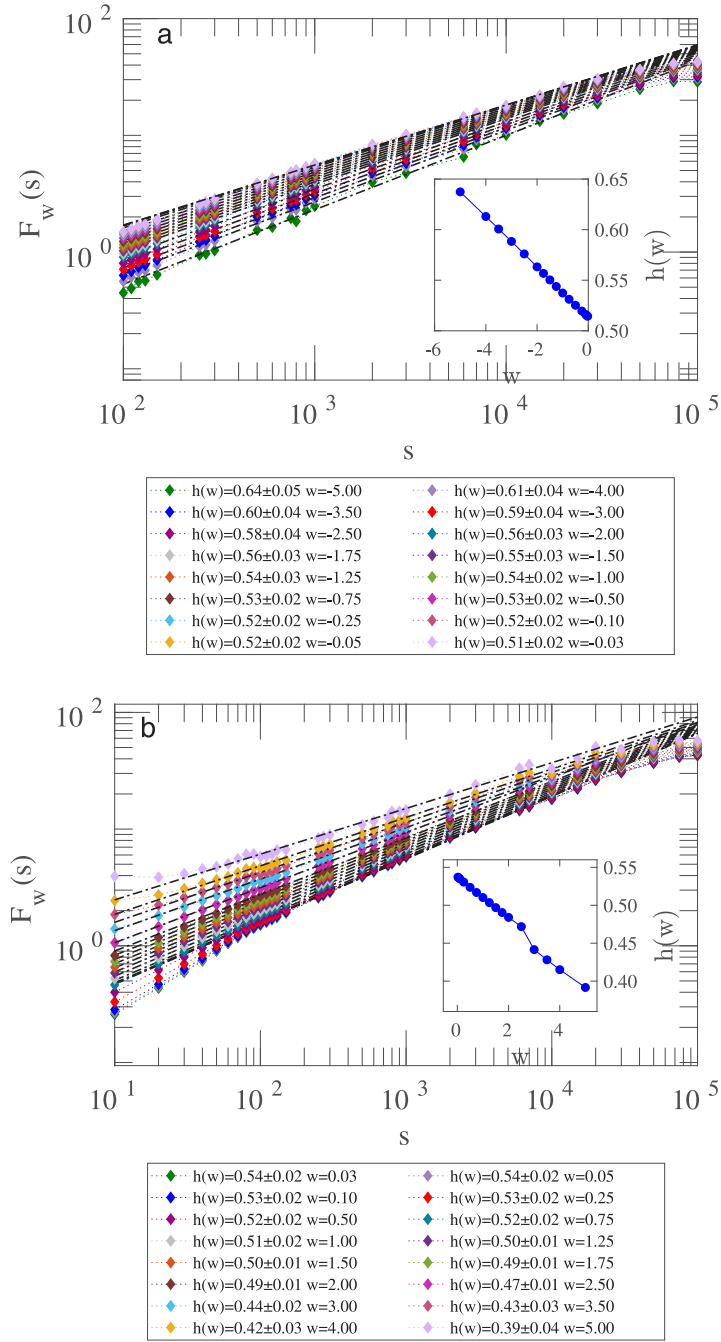


Fig. C.1. Detrended price return case: Calculation of the statistical function F_w using Eq. (C.2). The function of F_w vs s display power laws $F_w(s) \sim s^{h(w)}$, where $h(w)$ depend on w . This feature demonstrates that the time series is a multifractal.

Then, the Stationary Test is based on the autocorrelation and the power spectrum analysis of the time series is applied.

The Fourier Transformation (FT) of x_t is represented as $\hat{x}(f)$,

$$\hat{x}(f) = \frac{1}{\sqrt{T}} \sum_{t=0}^T x(t) e^{-2\pi i f t}. \quad (26)$$

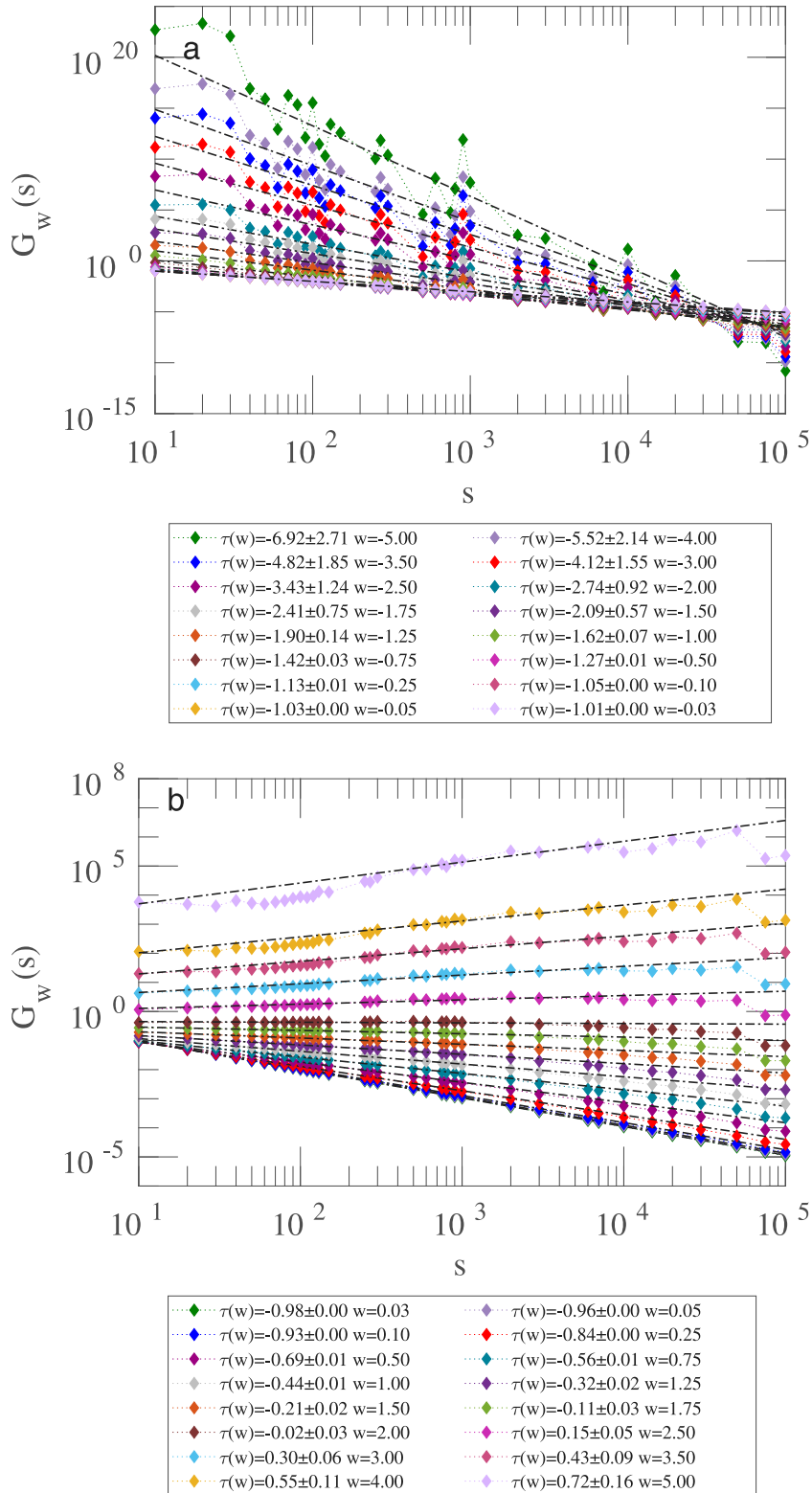


Fig. C.2. Detrended price return case: Calculation of the “generalized statistical functions” $G_w(s)$ vs s for negative (a) and positive (b) values of order w . The “generalized statistical functions” is calculated by applying Eq. (C.7).

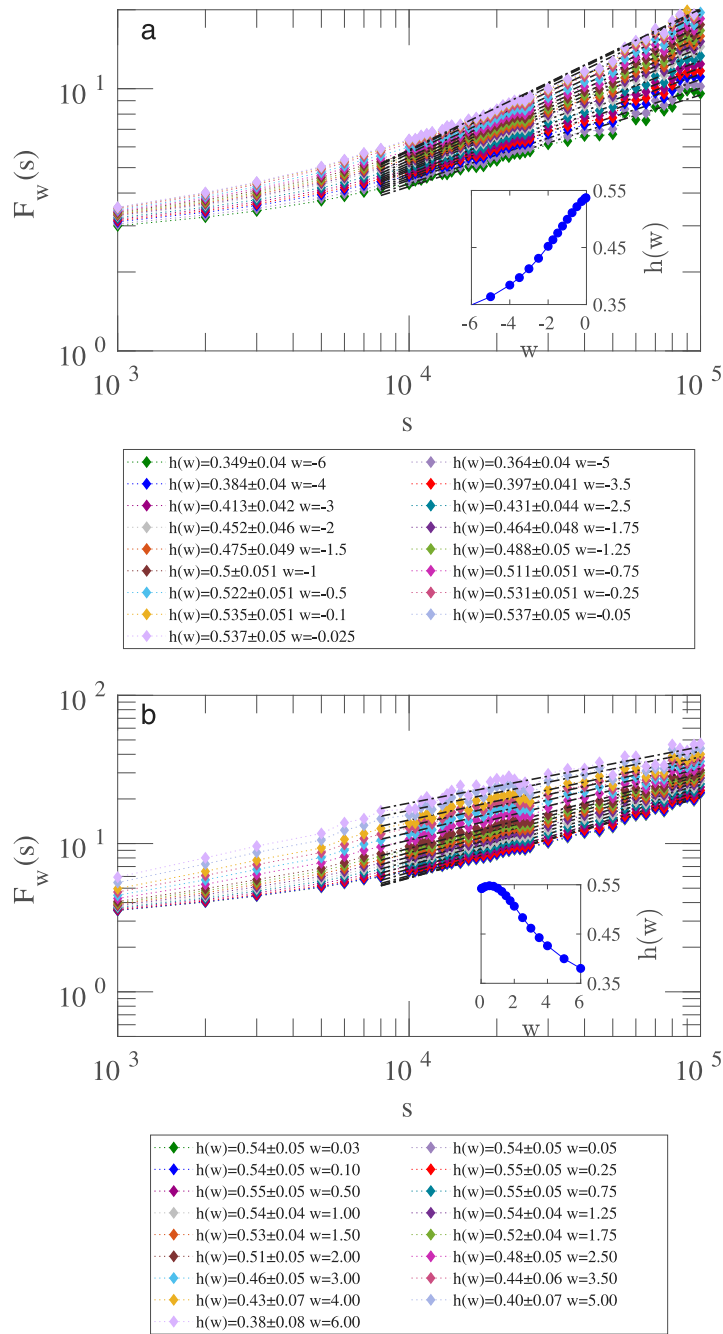


Fig. C.3. Normalized price return case: Calculation of the statistical function F_w using Eq. (C.2). The function of F_w vs s display power laws $F_w(s) \sim s^{h(w)}$, where $h(w)$ depend on w . This feature demonstrates that the time series is a multifractal.

This test states that any process is stationary if the power spectrum $|\hat{x}(f)|^2$ exists, and can be expressed as the Fourier transform of the autocorrelation $C(s)$ [36,69],

$$|\hat{x}(f)|^2 = \hat{C}_x(f), \quad (27)$$

We determine the detrended price return as a non-stationarity time series on account of the time-varying volatility and short time correlations. The normalized price return is stationarity based on the results shown in Fig. 7, where the match between $|\hat{x}^{**}(f)|^2$ and $\hat{C}_{x^{**}}(f)$ is achieved for time lags larger than 30 min. The $|\hat{x}^{**}(f)|^2$ and the $\hat{C}_{x^{**}}(f)$ have been smoothed by applying the Savitzky–Golay filtering. The smoothing is made with the purpose of reducing the noise of the data without

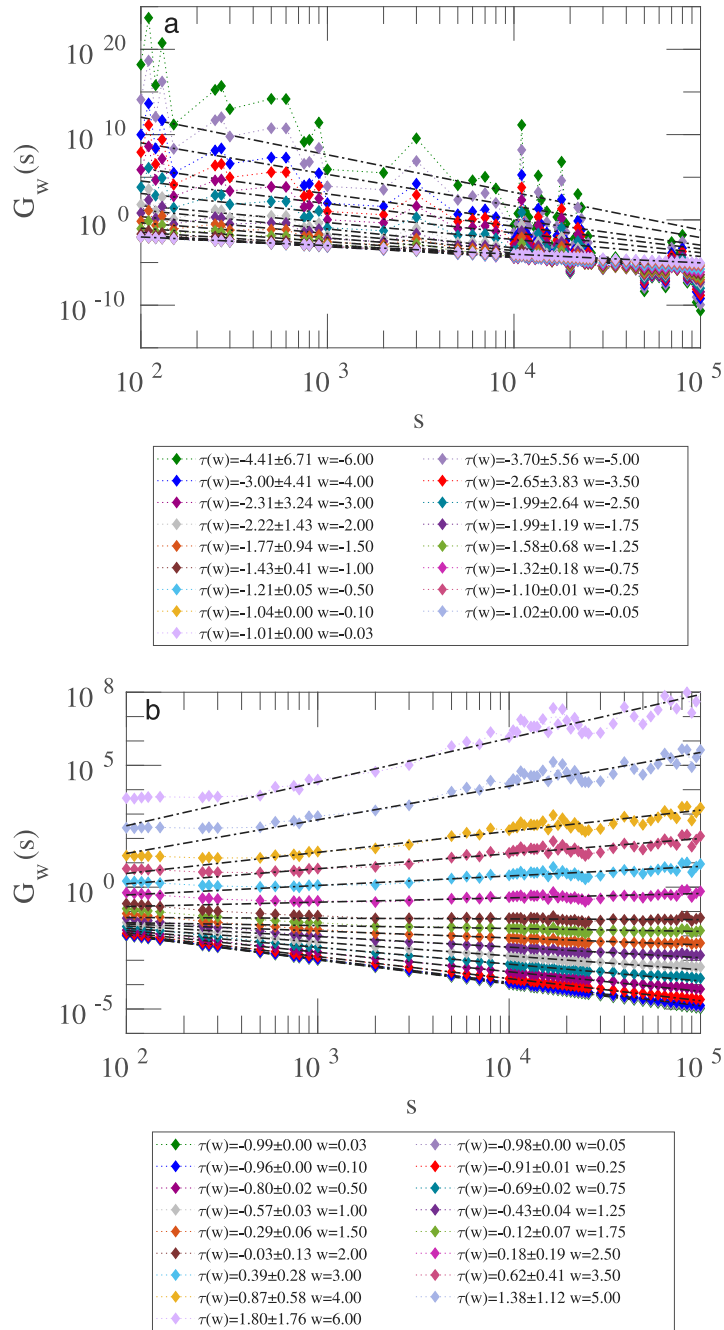


Fig. C.4. Normalized price return case: Calculation of the “generalized statistical functions” $G_w(s)$ vs s for negative (a) and positive (b) values of order w . The “generalized statistical functions” is calculated by applying Eq. (C.7).

distorting their tendency [70,71]. The smoothing obtained by the Savitzky–Golay filter is based on the least-squares of a linear fitting across a moving window by applying the normalized convolution integers of Savitzky–Golay, C_i [70,71]. For this case the length of the moving window is 3.97×10^{-5} Hz.

Also, the relation $|\hat{I}(f)|^2 = \hat{C}_I^2(f)$ is observed in Fig. 8, suggesting that the detrended index can be represented by a red noise [72] due to its inversely proportionality to f^2 .

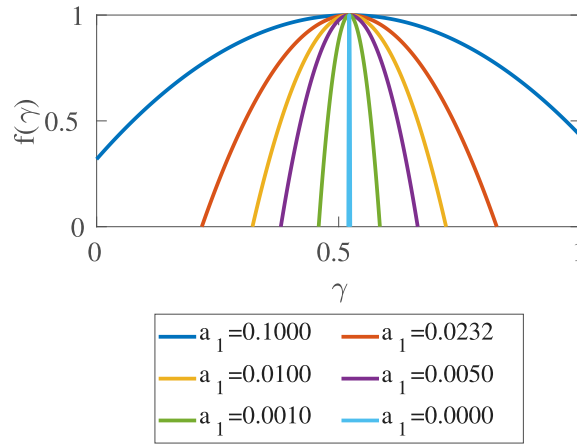


Fig. D.1. The range of probable generalized Hurst exponents values is reduced By making $a_1 \rightarrow 0$.

5. Scaling analysis of the fractal of detrended price return

For the scaling analysis, the Hurst exponents of the detrended and normalized price returns are calculated. The Hurst exponent is a measure of long-range memory of a time series. More specifically, it measures the statistical dependence of two points of the time series with respect to the time difference between them [33,73].

The Hurst exponent is better known as the *index of dependence*, and as such, it is a dimensionless estimator of self-similarity in a time series [5,74,75]. The self-similarity occurs when a structure repeats itself on subintervals, so they are *scale-invariant*, which is a property that some time series possess [5].

Two types of self-similar signals exist. The monofractal signal, which presents a unique scaling behavior of complexity. The scaling behavior is obtained from the ratio of the self-similar pattern that changes with the scale at which it is measured. The second type is the multifractal signal, which has more than one scaling behavior [33].

The Hurst exponent can be estimated by applying the detrended fluctuation analysis (DFA) [76,77] explained in Appendix C. The DFA is a technique used to demonstrate self-similarity in a time series [78], and the DFA has been applied widely in scientific studies to estimate the Hurst exponent [77]. Additionally, by analyzing the statistics, it is possible to characterize the time series as monofractal or multifractal if the self-similarity exists.

The first part of the DFA analysis consists in removing the trend of the original time series considering non-overlapping segments. However, our trend was removed previously in Section 3-B by applying a simple but still rigorous MA method considering overlapped time windows with an optimal size of time window t_w .

The second part of the DFA is based on the calculations of the *scaling function*, $F_w(s)$, as a function of the *time segment* s , where w is the mathematical moment's order (See Appendix C) [33,59,79]. If the *scaling function* of the time series is a power-law function and satisfies the $F_w(s) \sim s^H$, the time series is monofractal. Then, the exponent H is called the Hurst exponent [33,80]. Other time series do not have a unique, constant valued H . Their scaling exponent H depends on the order of the moment w (Fig. C.1). These are classified as a multifractal time series. The generalization method of DFA (G-DFA) is tailored for multifractal time series. In the G-DFA, the *generalized statistical moments* are defined by $G_w(s)$ in Eq. (C.7). If the series is multifractal, the exponent $\tau(w)$ is the so-called *generalized scaling exponent* (GSE). The multifractal time series satisfies the scaling law of $G_w(s) \sim s^{\tau(w)}$. For multifractal time series the generalized scaling exponent $\tau(w)$ can be defined in terms of the generalized Hurst exponent $h(w)$ [3,79]:

$$\tau(w) = wh(w) - 1. \quad (28)$$

Fig. 9-a shows the generalized Hurst exponent of the detrended price return obtained from Fig. C.1, and its fitting. The generalized Hurst exponent is approximated to $h(w) \simeq -0.023w + 0.521$ for $-5.0 \leq w \leq 5.0$. Fig. 9-a benefits for interpretation of the generalized Hurst exponent: $h(w) \in (0.5, 1)$, indicates persistence (correlation) of time series [81]. Fig. 9-a shows persistence of time series, when $w < 0$. This negative value of w represents the segments with small fluctuations [79]. On the other hand, the $h(w) \in (0, 0.5)$, indicates anti-persistence (no-correlation) of time series [81]. Fig. 9-a shows anti-persistence when $w > 0$, segments with large fluctuations [79].

Fig. 9-b shows the $\tau(w) = wh(w) - 1$ relationship, proving that the detrended price return is self-similar and multifractal [79]. Sub Fig. 9-c, shows the generalized Hurst exponent of the normalized price return obtained from Fig. C.3. The $h(w) \in (0, 0.5)$ indicates anti-persistence. Fig. 9-d proves that the normalized price return is self-similar and multifractal such as the detrended price return but with weaker correlation that rapidly decrease. Additional analysis about the behavior of the generalized Hurst exponent when it is linear, is displayed in Appendix D

6. Conclusions

We proposed an improved version of the Black–Scholes equations postulated in Section 2 of our paper, Eq (15). Our stochastic differential equation captures the index behavior more precisely by having a trend that fluctuates deterministically. The proposed SDE, Eq (15), presents its deterministic component as endogenous or exogenous predictive factors, which is associated with internal/external events that produce changes of trend direction. The stochastic part is represented by the q -Gaussian noise, which is a compound of a time-dependent standard deviation and Gaussian noise.

For the validation of our proposed SDE, the S&P500 index was detrended using a weighted average method with an overlapped time window of one year. The time window length was obtained by calculating an average Kurtosis equals three. The aim was to achieve an optimum window size such that for a given data set the distribution in each window is as close as possible to a Gaussian distribution. However, we showed that each distribution in each window has a high varying variance within an hour, and fits better with a q -Gaussian distribution function. For the authors, these results agree with Eq (15), where the detrended part is a q -Gaussian noise compounded of Gaussian noise and a variance term that depends on time (volatility clusters). By calculating an optimal window size, the detrended data set is symmetric, where extreme outlier events are improbable to occur.

This paper considered three methods to prove stationary. The first one was based on previous superstatistical models, determining optimal time window length to capture local Gaussians based on the Kurtosis application. The second method was based on the DFA, which proposes an alternative approach to test stationarity for monofractal time series after removing trends that cannot be normalized. The detrending price return is a multifractal time series. As a consequence, the DFA approach could not be used. The third method used was the Stationarity Test; this test is based on the power spectrum of the time series, which needs to be approximately equal to the Fourier transform of its autocorrelation. After the normalization of the detrended price, the Stationarity Test was applied. We obtained a match between the power spectrum and the Fourier transform of the autocorrelation after dismiss the first 30 min of the autocorrelations' time-lag. During these 30 min weak autocorrelations were detected on the normalized price return.

This paper does not provide an ultimate define proof that the normalized price return is perfectly stationary. The authors prefer to refer to a normalized price return as a weak stationary time series, where the mean and standard deviation fluctuates around a constant value. Future work should be focused on new methods that provide a clear distinction between a deterministic trend and non-stationary stochastic noise. Additionally, further research should focus on evaluating the relationship between short time-correlated time series and volatility clustered.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Cleansing data

The three artifacts removed from the S&P500 data are: In 17/06/1997, at the peak of the Asian financial crisis [55]. In 20/03/2000, when the imminent bankruptcy of many Internet companies was announced [56]. The third artifact removed was the Haiti earthquake that occurred on 11/01/2010 [57]. These three events introduce spurious jumps in the S&P500, producing a considerable percentage change of the Index for 1 to 8 min only. Then, the abrupt Index values return to the original. The three artifacts are shown in Fig. A.1(a–c) in chronological order.

Appendix B. Dickey–Fuller test

In statistics, the Dickey–Fuller analysis is the most popular test for determining the stationarity of time series [64,65]. The Dickey–Fuller test models autoregressive time series with random noise with the form:

$$I_t^* = \rho I_{t-1}^* + \epsilon_t, \quad (29)$$

where ϵ_t is a sequence of independent random variables, with mean zero following a normal distribution function. Fig. B.1 displays the linear relationship between I_t^* and I_{t-1}^* , where $\rho \approx 1$ and ϵ obeys a q -Gaussian distribution. As consequence, the Dickey–Fuller analysis will not be applicable for the stationarity test of detrended price return.

Appendix C. Detrended fluctuation analysis (DFA)

The Detrended Fluctuation Analysis method has become a widespread technique to determine the self-similarity of a time series based on statistical functions F_w [4,33]. The statistical functions F_w [Eq. (C.2)] are calculated based on central moments of the time series. For that, the detrended time series is divided into equal non-overlapping segments with a length of s . The s value represents the time segment in which the detrended time series will be analyzed.

Then, the Hurst exponent is obtained from the power law given by F_w vs s . [33,80].

C.1. Description of the method of DFA

We will proceed to indicate the steps to adopt this method to find the Hurst exponent of the detrended price return $x(t)$. First, we focus on calculating the statistical functions F_w , then to the calculation of Hurst exponent H [33,36,74,75,80].

- *Step 1:* The ‘profile’ is the cumulative sum of the time series to analyze. For this case, our profile matches with the detrended price $I^*(t)$ obtained after applying Eq. (19).
- *Step 2:* The profile $I^*(t)$ is divided into non-overlapping segments $[N_s = N/s]$ with the same length s .
- *Step 3:* The variance for each of the segments $v = 1, 2, 3, \dots, N_s$ is obtained by applying the following equation,

$$F^2(v, s) = \frac{1}{s} \sum_{i=1}^s (I^*[(v-1)s + i] - \bar{I}^*(v, s))^2, \quad (C.1)$$

where, $\bar{I}^*(v, s)$ represents the mean of each segment of $I^*(t)$.

- *Step 4:* The statistical moments are obtained considering a range of values for the w th orders.

$$F_w(s) = \left\{ \frac{1}{N_s} \sum_{i=1}^{N_s} [F^2(v, s)]^{w/2} \right\}^{1/w} \quad (C.2)$$

For detrended time series that present long-range correlations, the following power law is obtained [33,73,74,79]:

$$F_w(s) \sim s^{h(w)}. \quad (C.3)$$

For monofractal time series $h(w)$ is independent from w value due to a constant scaling behavior over all the segments ($F^2(v, s)$ in Eq. (C.1) is cte.) [14,33], so $H = h(w)$. Fig. C.1 shows the results of DFA analysis, where a power law between F_w vs. s is observed. However, the power laws depend on the w^{th} order. Consequently, the Multifractal Detrending Fluctuation Analysis (MF-DFA) is applied to evaluate the dependence between the power laws and the order degree w^{th} .

For a standard multifractal analysis (MF-DFA) the scaling exponent $\tau(w)$ needs to be obtained. This exponent is the power law defined between the $G_w(s)$ vs s . The $G_w(s)$ is a general statistical moment based on the following relationship:

$$G_w(s) = \sum_{v=1}^{N/s} |I^*(vs) - I^*((v-1)s)|^w, \quad (C.4)$$

where the term $I^*(vs) - I^*((v-1)s)$ is equal to the cumulative summation of the $x(t)$ within each segment v of size s .

$$\sum_{t=(v-1)s+1}^{vs} x_t = I^*(vs) - I^*((v-1)s). \quad (C.5)$$

In the standard multifractal formalism this sum is better known as the “box probability” $p_s(v)$, and it is used to find hypothetical probability values on the original profile $I^*(t)$.

$$p_s(v) = \sum_{t=(v-1)s+1}^{vs} x_t. \quad (C.6)$$

Then, by replacing Eq. (C.6) in Eq. (C.5). The “box probability” is defined in terms of the original profile, $p_s(v) = I^*(vs) - I^*((v-1)s)$. After that, we replace this equivalence in Eq. (C.4), the following relation is obtained:

$$G_w(s) = \sum_{v=1}^{N/s} |p_s(v)|^w, \quad (C.7)$$

where,

$$G_w(s) \sim s^{\tau(w)}. \quad (C.8)$$

The $\tau(w)$ represents the generalized scaling exponent (GSE), and is related with $h(w)$ for stationary and normalized time series [14,33],

$$\tau(w) = wh(w) - 1. \quad (C.9)$$

The Fig. 9, shows a summary of the scaling exponents $h(w)$ and $\tau(w)$, respectively. The relationship between $h(w)$ and $\tau(w)$ is noticed, proving that $x(t)$ is a multifractal stationarity time series. The Figs. C.2a and C.2b show the results of the generalized statistical function G_w for negative and positive values of w respectively.

Appendix D. Hurst exponent spectrum

The Legendre transformation is used to calculate the spectrum of the characteristic exponents after applying the detrending fluctuation analysis. We did the Legendre transformation of GSE as a function of $h(w)$. For the detrended price return the relationship between GSE and $h(w)$ is linear as can be observed in Fig. 9a. The Legendre transform of the GSE is:

$$f(\gamma) = \gamma w - \tau(w) \quad (D.1)$$

where,

$$\gamma = \frac{\partial \tau(w)}{\partial w} = h(w) - w \frac{\partial h}{\partial w}$$

By replacing the Eq. (C.9) in Eq. (D.1), the Legendre transform of $\tau(w)$ is:

$$f(\gamma) = w(\gamma - h(w)) + 1 \quad (D.2)$$

For monofractal time series $h(w) = cte$, implying that $\gamma = H$. On the other hand, for multifractal time series $h(w) \neq cte$, then:

$$\begin{array}{ll} \text{monofractal} & f(\gamma) = 1 \quad \gamma = H \\ \text{multifractal} & f(\gamma) = 0 \quad \gamma \neq cte \end{array} \quad (D.3)$$

From Fig. 9a, the generalized Hurst exponent is linear $h(w) = a_1 w + b_1$. As a consequence the value of $\gamma = b_1 - 2a_1 w$, and

$$w = \frac{b_1 - \gamma}{2a_1} \quad (D.4)$$

By replacing Eq. (D.4) in Eq. (D.2) the following relation is obtained:

$$f(\gamma) = 1 - \frac{1}{4a_1}(b_1 - \gamma)^2 \quad (D.5)$$

The Fig. D.1, represents how the range of probable generalized Hurst exponents values is reduced from multifractal time series to monofractal time series. At the limit of $a_1 \rightarrow 0$, one finds that $h(w) = H$ retrieving a Dirac delta function shifted representing a monofractal time series.

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